EXPERIMENT NO. 8

Implement Inheritance and the super Keyword

PART B – TO BE COMPLETED BY STUDENTS

Roll No. C-60

Name: Avdhut Parish Madhavi

Class / Division: SE-C

Batch: C2

Date of Experiment _____

Date of Submission _____

Grade _____

B.1 Document Created by the Student

Create another subclass representing a different healthcare role and demonstrate the inheritance chain.

An additional subclass named Nurse is created to represent another healthcare role. It extends the Doctor class, which already extends HospitalStaff, demonstrating multilevel inheritance. The super keyword is used to invoke parent constructors and methods.

B.2 Student Program / Output

Paste/write the final program and attach or reproduce the observed output below.

```
class HospitalStaff {
    String name;
    HospitalStaff(String name) {
        this.name = name;
    }
    void displayRole() {
        System.out.println("Hospital Staff: " + name);
    }
}
```

```
class Doctor extends HospitalStaff {
    String department;
    Doctor(String name, String department) {
        super(name);
        this.department = department;
    }
    void displayDoctor() {
        super.displayRole();
        System.out.println("Department: " + department);
    }
}
class Nurse extends Doctor {
    String duty;
    Nurse(String name, String department, String duty) {
        super(name, department);
        this.duty = duty;
    }
    void displayNurse() {
        displayDoctor();
        System.out.println("Duty: " + duty);
    }
}
public class InheritanceDemo {
    public static void main(String[] args) {
        Nurse n = new Nurse(
            "Priya", "Emergency", "Patient Care"
        );
        n.displayNurse();
```

```
    }
}
```

```
Hospital Staff: Priya
Department: Emergency
Duty: Patient Care
```

B.3 Observations and Learning

● What is code reusability in inheritance?

Code reusability means using the fields and methods of an existing class in another class without rewriting the same code.

● What does super() do?

super() calls the constructor of the immediate parent class. The super keyword can also access parent-class methods and variables.

● How does multilevel inheritance differ from single inheritance?

Single inheritance involves one parent and one child class. Multilevel inheritance forms a chain in which one class inherits from another derived class.

Student observation / learning:

I learned how inheritance promotes code reuse and how super() calls parent constructors. The program demonstrates multilevel inheritance through HospitalStaff, Doctor, and Nurse.

B.4 Conclusion

Healthcare staff hierarchy is directly consistent with the syllabus's healthcare-context activities.

Student conclusion:

The experiment successfully demonstrates inheritance, code reusability, and the use of the super keyword. Multilevel inheritance represents a hierarchy of related healthcare roles.

B.5 Questions of Curiosity

6. What is code reusability in inheritance?

It is the ability to reuse existing class properties and methods in derived classes.

7. What does super() do?

It invokes the immediate parent class constructor and helps initialize inherited members.

8. How does multilevel inheritance work?

Multilevel inheritance occurs when a class inherits from a derived class, forming a chain such as HospitalStaff → Doctor → Nurse.